higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(Second Paper)
NQF LEVEL 4

7 November 2019

This marking guideline consists of 13 pages.

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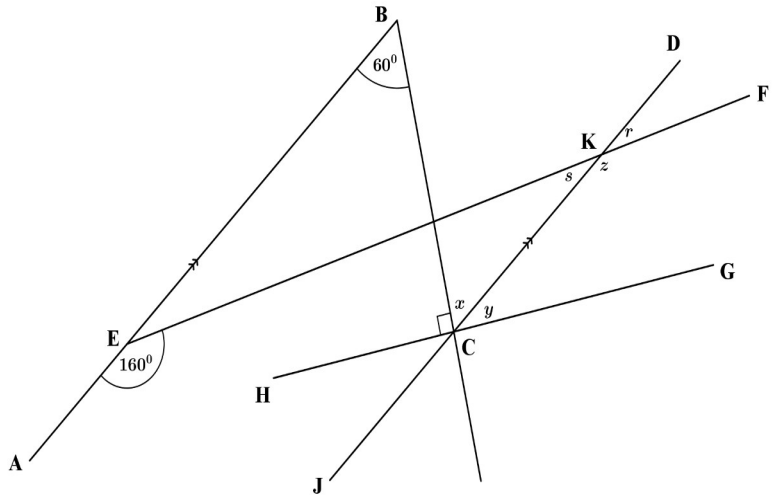
✓ = 1 mark
√ = ½ mark

- Accept ALL mathematically valid solutions.
- Consistent accuracy will apply as a general rule.
- If a candidate does a question twice and does NOT delete either, mark the FIRST attempt.
- If a candidate does a question, crosses it out and does not redo it, mark the deleted attempt.

MARKING LEGEND	
SYMBOL	**EXPLANATION OF SYMBOLS**
✓A	Accuracy
✓M	Method
✓CA	Continuous accuracy (follow-up)
✓B/D	Break down
✓S	Statement
✓R	Reason
✓S/R	Statement and Reason

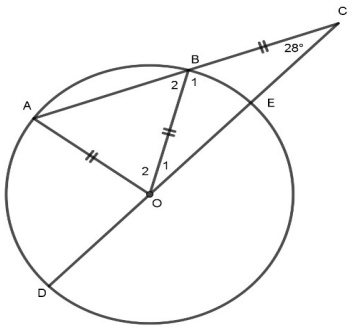
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QUESTION 1



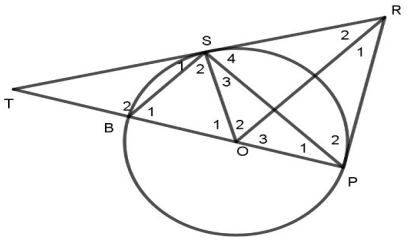
	SOLUTION	EXPLANATION		
1.1	$x = 60°$	1 for answer only	(A)	
	$y = 90 - x = 90 - 60$ $= 30°$	1 for answer	(CA)	
	$s = 180° - 160°$ (supp. angles) $= 20°$	1 for answer only	(A)	
	$r = 20°$ $(vertically\ \ opposite\ \ angles)$	1 for answer	(CA)	
	$z = 160°$ (straight angle)	1 for answer	(CA)	(5)

1.2

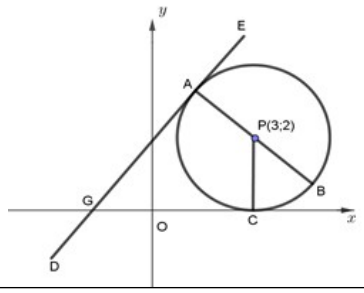


$AO = OB$ (Radii) $BC = OA$ (Given) $\therefore OB = BC$ √ $\hat{O}_1 = \hat{C}$ $\hat{O}_1 = 28°$ √ (Angles opp. equal sides)√ *But* $\hat{B}_2 = \hat{O}_1 + \hat{C}$ (Ext. ∠'s angles of Δ) $= 28° + 28° = 56°$ ¿ $\hat{A} = \hat{B}_2$ (Opp. radii)√ $\therefore \hat{A} = 56°$ ✓ ¿ $A\hat{O}D = \hat{A} + \hat{C}$ (Ext. ∠ of Δ) $= 56° + 28°$ $= 84°$	½ for OB = OC (A) ½ for answer ½ for reason (A) 1 for answer (A) ½ for reason (A) 1 for angle A (A) 1 for answer	(5)

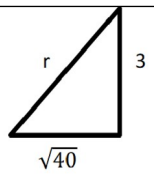
1.3



1.3.1	$\hat{P}_1 + \hat{P}_2 = 90°$ √ (Tangent radius)√ $\hat{S}_1 + \hat{S}_2 = 90°$ √ (Tangent radius)√ $\hat{P}_1 + \hat{P}_2 = \hat{S}_1 + \hat{S}_2$ $\therefore POSR$ is a cyclic quad. (Ext. ∠ of cq = opp. ∫.∠	½ for S, ½ for R ½ for S, ½ for R 1 for R	(3)
1.3.2	$\hat{S}_1 = \hat{P}_1 = x$ √ (Tangent chord) $\hat{P}_1 = \hat{R}_2 = x$ √(Subtended by SO) from QUESTION1.3.1 cyclic quadrilateral $\therefore S_1 = \hat{R}_2$ (¿ statement above) $\therefore SB \parallel \vee$ (Corr. angles)	½ for S ½ for S 1 for R	(2)

1.4				
	1.4.1	*The radius is perpendicular ¿ the tangent.*	1 for answer (A)	(1)
	1.4.2	$(3;0)$	½ per coordinate (A)	(1)
	1.4.3	$(x-a)^2+(y-b)^2=r^2\surd$ $P(3;2)$: $(x-3)^2+(y-2)^2=r^2\surd$ $PC=r=2-0=2$ $(x-3)^2+(y-2)^2=2^2$ $(x-3)^2+(y-2)^2=4$	½ for formula (A) ½ for substituting centre (A) 1 for radius (A) 1 for answer (CA)	(3)
	1.4.4	*Equation of tangent given*: $y=\frac{4}{3}x+\frac{4}{3}$ $y=0$ $0=\frac{4}{3}+\frac{4}{3}$ $x=-\frac{4}{3}\times\frac{3}{4}$ $x=-1$ $(-1;0)$	1 for setting $y=0$ (A) ½ for each coordinate (CA)	(2)
	1.4.5	$AB\perp\geq$ (tan *chord*) $m_{AB}\times m_{GE}=-1$ $m_{AB}\times\frac{4}{3}=-1$ $m_{AB}=-\frac{3}{4}$ ✓ $y-y_1=m(x-x_1)$ $y-y_1=-\frac{3}{4}(x-x_1)$ $P(3;2)$ $y-2=-\frac{3}{4}(x-3)$ $y=-\frac{3}{4}x+\frac{9}{4}+2$	1 for *m* of AB (A) 1 for substituting point (A) 1 for answer (CA)	(3)

		$y=-\frac{3}{4}x+\frac{17}{4}$		
				[25]

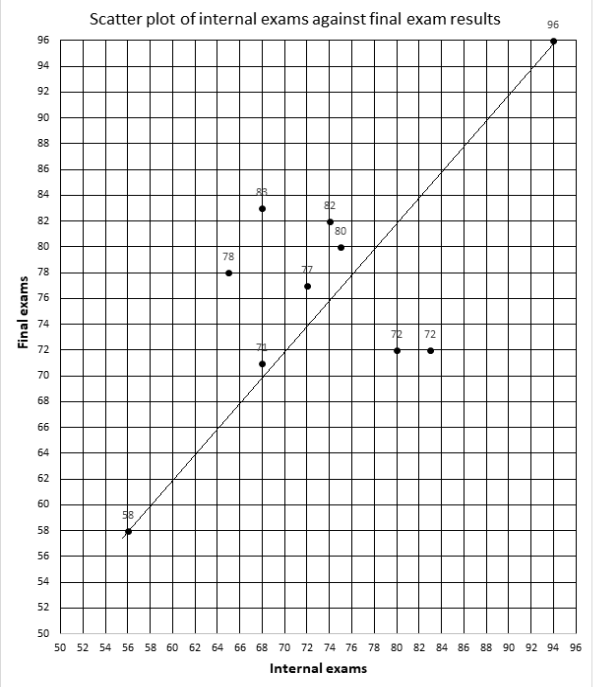
QUESTION 2			
2.1	$\sin 290°$ $\neq \sin(360°-70°)$ $=-\sin 70°$	1 for answer (A)	(1)
2.2	 $\tan x=\frac{3}{\sqrt{40}}$ $r=7$ ✓ … by Pythagoras $\cos x=\frac{\sqrt{40}}{7}$ ✓	1 for r (A) 1 for answer (CA)	(2)
2.3	$\frac{\sin 2x\cos x+\cos 2x\sin x}{\cos 2x\cos x-\sin 2x\sin x}=1$ $\frac{\sin(2x+x)\surd}{\cos(2x+x)\surd}=1 \tan 3x=1$✓ $3x=\tan^{-1} 1$ $3x=45^0$✓ $x=15^0\surd;\ 75^0\surd$	½ each for num and denomin (A) 1 for $\tan 3x$ (A) 1 for ref angle (A) ½ for each answer (A)	(4)
2.4	$\frac{2\sin 50°(\cos 60°\cos 10°+\sin 60°\sin 10°)}{\cos 10°}$ $=\frac{2\sin 50°\cos(60°-10°)}{\cos 10°}$ $=\frac{2\sin 50°\cos 50°}{\cos 10°}$ ¿ $\frac{\sin 2.50°}{\cos 10°}$ ¿ $\frac{\sin 100°}{\cos 10°}$ ¿ $\frac{\sin(90°+10°)}{\cos 10°}$	1 for identifying $\cos(A-B)$(A) 1 for applying double angle (A)	(4)

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	¿ $\frac{\cos 10° \checkmark}{\cos 10°}$ $=1$	1 for applying $\sin(90^0 + x)$ (A) 1 for answer (A)	

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2.5	2.5.1	$\cos(x+y)=\cos x\cos y-\sin x\sin y$	1 for answer (A)	(1)
	2.5.2	$\cos 2x=\cos(x+x)\surd$ ¿ $\cos x\cos x-\sin x\sin x\surd$ ¿ $\cos^2 x-\sin^2 x$ $\surd$ $=1-\sin^2 x-\sin^2 x$ $=1-2\sin^2 x\surd$	1 for splitting $2x$ (A) ½ for expanding (A) ½ for simplifying (A) 1 for replacing $\cos^2 x$ (A) ½ for arrive at answer (A)	(3)
2.6	$\frac{\cos x+\sin x}{\cos x-\sin x}-\frac{\cos x-\sin x}{\cos x+\sin x}=2\tan 2x$ $LHS=\frac{\cos x+\sin x}{\cos x-\sin x}-\frac{\cos x-\sin x}{\cos x+\sin x}$ ¿ $\frac{(\cos x+\sin x)^2-(\cos x-\sin x)^2}{(\cos x-\sin x)(\cos x+\sin x)}$ ¿ $\frac{1+2\sin x\ \cos x-(1-2\sin x\ \cos x)\surd}{\cos^2 x-\sin^2 x\ \surd}$ ¿ $\frac{4\ \sin x\ \cos x}{\cos 2x}$ ¿ $\frac{2\ \sin 2x}{\cos 2x}=2\ \tan 2x=RHS$	1 for adding correctly (A) ½ for numerator (A) ½ for denominator (A) 1 for denominator (A) 1 for numerator (A) 1 for arriving at answer (A)	(5)
2.7	2.7.1	$A\hat{B}C=180°-(43+36)\quad=101°$	1 for answer (A)	(1)
	2.7.2	$\frac{sinA}{a}=\frac{sinB}{b}\surd\frac{\sin 43°}{BC}=\frac{\sin 101°}{2\ 000}$ $BC=1\ 389{,}52\ m\ BC=1\ 389{,}52\ m$ $BC=DF\frac{DF}{DE}=\cos 27°\ \surd$ $DE=\frac{1\ 389{,}52}{\cos 27°}$ DE $=1\ 559{,}5\ m$	½ for formula (A) 1 for correct substitution (A) 1 for BC (CA) ½ for ratio (A) 1 for answer (CA)	(4)
				[25]

QUESTION 3		
3.1 and 3.3	 *Plotting points*: *All* 10 *points correct* - 4 *marks* 6 - 9 *points correct* - 3 *marks* 3 - 5 *points correct* - 2 *marks*	(4)
3.3	*Line of best fit*	(2)

3.2 (OPTION 1)

Internal examinations (x)	Final examination (y)	xy	x^2
80	72	5760	6,400
68	71	4,828	4,624
94	96	9,024	8,836
72	77	5,544	5,184
74	82	6,068	5,476
83	72	5,976	6,889
56	58	3,248	3,136
68	83	5,644	4,624
65	78	5,070	4,225
75	80	6,000	5,625
$\sum x = 735\surd$	$\sum y = 769\surd$	$\sum x.y = 57\ 162\surd$	$\sum x^2 = 55\ 019\surd$
$\bar{x} = 73{,}5\ \surd$	$\bar{y} = 76{,}9\surd$		

Use ticks as indicated on table.

$$b = \frac{n\sum x.y - \sum x \sum y}{n\sum x^2 - \left(\sum x\right)^2}$$

$$= \frac{10 \times 57\ 162 - (735)(769)}{10 \times 55\ 019 - (735)^2}$$

$$= \frac{571\ 620 - 565\ 215}{550\ 190 - 540\ 225} \qquad = \frac{6\ 405}{9\ 965}$$

$b = 0{,}64$

$a = \bar{y} - b\bar{x}$

$a = 76{,}9 - 0{,}64(73{,}5)$

$a = 29{,}66$

$\therefore y = a + bx$

$y = 29{,}66 + 0{,}64\ x$

1 for substitution in formula (CA)

1 for value of a (CA)

1 for equation (CA) (6)

3.2 (OPTION 2)

Internal examinations (x)	Final examination (y)	$(x-\bar{x})(y-\bar{y})$	$(x-\bar{x})^2$
80	72	-31,85	042,25
68	71	32,45	30,25
94	96	391,55	420,25
72	77	-0,15	2,25
74	82	2,55	0,25
83	72	-46,55	90,25
56	58	330,75	306,25
68	83	-33,55	30,25
65	78	-9,35	72,25
75	80	4,65	2,25
$\sum x = 735$	$\sum = 769$	$\Sigma(x-\bar{x})(y-\bar{y}) = 640,5$	$\left(\sum x-\bar{x}\right)^2 = 996.5$
$\bar{x} = 73,5\surd$	$\bar{y} = 76,9\surd$		

Use ticks as indicated on table.

$$b = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

$b = \frac{640,5}{996,5}\surd b = 0,64\surd$

½ for substitution in formula (CA)

1 for value of b (CA)

¿ $\bar{y} - b\bar{x}$

$a = 76,9 - 0,64(73,5)$

$a = 29,66$

1 for value of a (CA)

$\therefore y = a + bx$

$y = 29,66 + 0,64\ x$

1 for equation (CA)

3.4

$y = 29,66 + 0,64\ x$

$y = 29,66 + 0,64(70)$

$= 74,46$

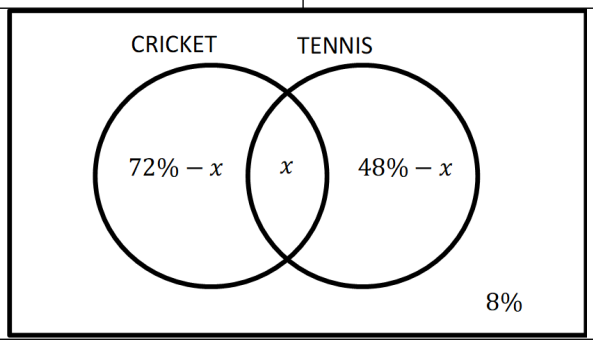
1 for formula (CA)

1 for substitution (CA)

1 for answer (CA)

(3)

[15]

QUESTION 4				
4.1	4.1.1	$100\%-8\%=92\%$ $\therefore 92\%\ of\ 700$ ¿ $\frac{92}{100}\times 700$ $=644\ learners$	Allocate according to ticks	
	4.1.2			

		$100\%=72\%-x+\ x+48\%-x+8\%$ $100\%+\ x=72\%+\ 48\%+8\%\qquad x=128\%-100\%$ $x=28\%$ $Only\ \ one\ \ facility=(72\%-x)+(48\%-x)$ $Only\ \ one\ \ facility=(72\%-28\%)+(48\%-28\%)=64\%$ $=0,64$	1 for equation (A) 1 for x (A) 1 for answer (A)	
	4.1.3	$P(using\ \ cricket\ \ nets)\times P(using\ \ tennis\ \ courts)$ $=0,72\times 0,48$ $=0,3456$ $P[(using\ \ cricket\ \ nets)\ \wedge\ P(using\ \ tennis\ \ courts)]$ $=0,28\surd$ $\therefore P(C\ \wedge\ T)\neq P(C\ )P(T)\surd$ $\therefore not\ \ independent$	Allocate according to ticks	
			(3 × 3)	(9)
4.2	4.2.1	$Receiving\ \ financial\ \ aid=\frac{6\ \ 106}{10\ \ 730}=0,57$	Allocate according to ticks	
	4.2.2	$P(repeater\ \ not\ \ receiving\ \ financial\ \ aid)$ ¿ $\frac{731}{10\ \ 730}=0,068$	Allocate according to ticks	

	4.2.3	$P(\text{first - receiving financial aid})$ ¿ $\frac{4\ 222}{10\ 730}$ $=0,39$	Allocate according to ticks	
			(3 × 2)	(6)
				[15]

QUESTION 5				
5.1	$A=P(1+i)^n \quad =R3\ 000\ 000\ (1+0,06)^6 \quad =R4\ 255\ 557,34$ $Dhimbi's\ investment = R4\ 255\ 557,34\ after\ six\ years$		Allocate according to ticks	(3)
5.2	5.2.1	*DEPOSIT* $10\%\ of\ R210\ 000 = R21\ 000\ \surd$ $Balance\ due = R210\ 000 - R21\ 000$ $= R189\ 000\ \surd$ *LOAN AMOUNT* $A=P(1+i\times n)$ $=R189\ 000(1+0,14\times 3)$ $=189\ 000(1,42)$ $=R268\ 380$ *Calculate monthly repayments* $Monthly\ repayments = \frac{268\ 380}{36}$ $= R7\ 455$ $\therefore Total\ repayments\ including\ insurance$ $= R7\ 455 + R22$ $=R7\ 477$	Allocate according to ticks	(6)

	5.2.2	*Total cost ¿ Weed ∧ Lawn* $R7\ 477 \times 36 = 269\ 172$ $\therefore Total\ cost = 269\ 172 + deposit$ $= R269\ 172 + R21\ 000$ $= 290\ 172$ *Therefore total cost* $= 290\ 172$	Allocate according to ticks	(3)
5.3	5.3.1	*GROSS INCOME* *Income ¿ salary* $= R25\ 000 \times 12 = R300\ 000$ *Investment* $= R1\ 000 \times 12$ $= R12\ 000$ *Bonus* $(once\ yearly) = R30\ 000$ $= R30\ 000$ $TOTAL = \underline{R342\ 000}$ √ *Deductions* *Pension* $= R2\ 500 \times 12$ $= R30\ 000$ *UIF* $= R250 \times 12$ $= R3\ 000$√ $\therefore Annual\ taxable\ income$ $= Gross\ income - (Pension + UIF)$ $= R342\ 000 - (R30\ 000 + R3\ 000)$ $= R309\ 000$√	Allocate according to ticks	(2)
	5.3.2	305 851– 423 300 63 853 + 31% of taxable income above 305 850	1 for answer (CA)	(1)

	5.3.3	$Tax = R\,63\ \ 853 + \frac{31}{100}(R\,309\ \ 000 - R\,305\ \ 850)$ $= R\,64\ \ 829{,}50$ *LESS REBATES* AND TAX CREDITS *Primary* √ *Medical aid tax credit*: $R\,310 \times 12$ = $R\,3\ \ 720$ √ *Medical 1st depend*: $R\,310 \times 12$ = $R\,3\ \ 720$ √ *Medical child*: $R\,209 \times 12 \times 2$ = $R\,5\ \ 016$ √ TOTAL = R26 523 √ Total tax due = payable tax - (rebates + tax credits) = $R\,64\ \ 829{,}50$ - R26 523 = R38 306,50 √ Total tax already paid = R2 500 × 12 = R30 000 Due by Mr Dedricks = payable tax − tax paid = R38 306,50 - R30 000 = R8 306,50	Allocate according to ticks	(5)
				[25]
			TOTAL:	**100**

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